

Math 61: Introduction to Discrete Structures

Discussion 6: Intro to Proofs 2/13/24, 2/15/24

1 Counting

Tip: for problems involving counting, always try to find a way to rephrase your problem as “selecting a subset (of chosen spots out of total number of spots)” (combinations/binomial coefficients $\binom{n}{k}$) or “choosing a sequence element by element (perhaps from different sets each time)” (permutations $\frac{n!}{(n-k)!}$ if choosing without replacement; or using multiplication rule more generally). Sometimes it can be subtle.

1. Choosing a 10-character password from set of 26 letters is clearly **choosing a sequence (one element after the next)** (one character at a time)
2. Counting 5-card hands drawn from a deck of 52 cards is clearly **counting subsets** (when the cards are in your hands, you don't care about order, just on the set of cards you have)
3. Counting the number of sequences of 10 coin flips is of the **choosing a sequence (one element after the next)**: one has 2 choices for the first, 2nd, 3rd, and so forth for a total of 2^{10} .
4. Counting the number of sequences of 10 coin flips with exactly TWO heads, is not naturally a “choosing a sequence” type problem, since we'd have to say something like “1st element can be H or T; 2nd element can be H or T; but 3rd element must be T if the first two are H's but if not it can be H or T; ...”. It is more naturally a problem about **counting subsets** of 2 spots out of 10 total for the H's to reside in.

1.1 Counting

1. Consider the integer lattice $\mathbb{Z}^2 \subseteq \mathbb{R}^2$. How many paths are there starting at $(0,0)$, going to (n_1, n_2) (for $n_1, n_2 \geq 0$), where you're only allowed to move one unit to the right (positive x direction) or one unit up (positive y direction) at each step? (Hint: encode such a path by a string on an alphabet of 2 symbols, 'R' and 'U')
 - Connecting the integer lattice with horizontal and vertical gridlines, every step you're now allowed to walk along one such line for one unit. Starting at the origin, how long is the minimal-length path from $(0,0)$ to $(-5,-5)$? How many such minimal-length paths are there?
 - For the integer grid in 3D space $\mathbb{R}^3$, what's the length of a minimal-length path from $(0,0,0)$ to $(5,5,5)$? How many such paths are there? How many such paths that pass through $(2,3,4)$?
2. Define a *composition* $\vec{\lambda} \vdash n$ to be a **vector** (different orderings of the same numbers are different vectors!) $\vec{\lambda} = (\lambda_1, \dots, \lambda_k)$ where $\lambda_1 + \dots + \lambda_k = n$ and $\lambda_i \in \mathbb{N}^+$, and k is the length of $\vec{\lambda}$, denoted $k = |\vec{\lambda}|$. For example for $n = 3$, the compositions of n are $(1, 1, 1), (2, 1), (1, 2), (3)$.
 - List out all compositions $\vec{\lambda} \vdash 4$ and 5.
 - How many compositions are there for each $n \in \mathbb{N}^+$? (Hint: consider groupings of a list of n 1's)
 - How many compositions $\vec{\lambda} \vdash 15$ are there of length $|\vec{\lambda}| = 3$?
 - How many compositions $\vec{\lambda} \vdash 15$ are there of length $|\vec{\lambda}| = 3$ with $\lambda_1 \geq 10$? What about with $\lambda_2 \geq 10$?

- How many compositions $\vec{\lambda} \vdash 15$ are there of length $|\vec{\lambda}| = 3$, where all $\lambda_1, \lambda_2, \lambda_3$ are single digit numbers?
- How many compositions $\vec{\lambda} \vdash 25$ are there of length $|\vec{\lambda}| = 4$?
 - How many compositions $\vec{\lambda} \vdash 25$ are there of length $|\vec{\lambda}| = 4$ with $\lambda_1 \geq 10$? What about with (at least) one of the λ_i ($i = 1, 2, 3, 4$) ≥ 10 ?
 - How many compositions $\vec{\lambda} \vdash 25$ are there of length $|\vec{\lambda}| = 4$ with BOTH $\lambda_1, \lambda_2 \geq 10$? What about with any 2 of the λ_i ($i = 1, 2, 3, 4$) ≥ 10 ?
 - How many compositions $\vec{\lambda} \vdash 25$ are there of length $|\vec{\lambda}| = 4$, where all $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are single digit numbers?

1.2 Pigeonhole

1. (Yes, I'm legally obligated to put example here) Here's some trivia: the average person has about 100,000 hairs on their head at one time. Let's assume that nobody has more than 200,000 hairs on their head. Suppose that we are in a city with more than a million inhabitants (e.g. Los Angeles). Show that there are at least $k = 2$ people with *exactly!* the same number of hairs on their head. Is it true with $k = 3$, or $k = 4$?
2. (Source) Show that some number in the sequence $1, 11, 111, 1111, \dots$ is divisible by 17
3. Given a sequence of n integers $(a_1, \dots, a_n)$, show that there is a consecutive subsequence whose sum is divisible by n . For example, given the sequence $(2, 7, 3, 19)$ we have the consecutive sum $2 + 7 + 3 = 12$ is divisible by 4, and $(3, 7, 3, 18)$ has $7 + 3 + 18 = 28$ is divisible by 4. (Hint: consider *initial* consecutive sums $I_k := a_1 + \dots, a_k$, and note that *differences* of these initial consecutive sums remain consecutive sums.)
4. By studying the long-division algorithm dividing p/q for $p, q \in \mathbb{N}^+$, show that the resulting decimal expansion you get either terminates or repeats after some point.
5. Given a real number x and natural number n , show that there are integers $k \in [1..N] := \{1, \dots, n\}, b \in \mathbb{N}$ s.t. $|kx - b| < \frac{1}{n}$. (Hint: for $k = 0, 1, \dots, n$, break down $kx = z_k + r_k$ into the integer part $z_k := \lfloor kx \rfloor$ and r_k the fractional part in $[0, 1)$. Then $r_k = kx - z_k$. You now have $n + 1$ "pigeons" $r_k \in [0, 1)$ and interval of length 1, and you're ultimately aiming for a bound of $< \frac{1}{n} \dots$ hmmm....)
 - Conclude that given any real number x , there are infinitely many (relatively prime) integers b, k s.t.

$$\left| x - \frac{b}{k} \right| < \frac{1}{k^2}.$$

6. For any $n \in \mathbb{N}^+$, consider the set of the first $2n$ positive integers, $[1..2n] := \{1, \dots, 2n\}$. Show that choosing any subset of size $n + 1$ from this set, there are 2 numbers that will be consecutive. (In particular, because consecutive numbers are relatively prime, this shows that any size $n + 1$ subset of $[1..2n]$ must contain 2 relatively prime numbers.)
7. (Dual to previous problem) Show that any size $n + 1$ subset of $[1..2n]$ must contain 2 numbers a, b s.t. $a \mid b$. One can approach via induction, or via the pigeonhole principle (the objects are definitely the $n + 1$ integers; but what are the n boxes? That's the difficult/creative part.)

1.3 Recurrences

1. A *full binary tree (FBT)* can be constructed as follows: start with a "root" node R , and apply the following procedure as many times as you want: choose a node with no children, and add 2 children below it, a *left child* and right child (forming a " $\wedge$ " shape). A node with no children is called a *leaf node* (note that a node can either have 0 children or 2 children).
 - Define B_n to be the number of full binary trees with n leaf nodes. For example, $B_1 = 1$ (the tree of just the root node), and $B_2 = 1$ (the tree that looks like " $\wedge$ "), and $B_3 = 2$ because from the 2-leaf tree " $\wedge$ ", one can add 2 children to the left child of the root, or add 2 children to the right child of the root. **Your turn:** calculate B_3, B_4, B_5 and draw each tree that you count.
 - Suppose now I tell you that $B_7 = 132$. Use this information to tell me how many 8-leaf FBTs there are, where the left child of the root has 0 children.

- For any FBT, we define its *left subFBT* (*left subtree for short*) as the left child of the root and all of its descendants; and similarly for the *right subFBT* (*right subtree for short*). Now use your knowledge of B_7 and B_3 to tell me how many 10-leaf FBTs there are, who have a 3-leaf left subtree and a 7-leaf right subtree.
- Can you come up with a recursive formula for B_{n+1} , in terms of $B_1, \dots, B_n$? (Hint: above you used the multiplication rule. To make this recursive formula, also use the addition rule.)

2. Recall from above the notation $\vec{\lambda} \vdash n$ for a *combination* of n . Let us define $R_1 = 1$ and

$$R_{n+1} := \sum_{\vec{\lambda} \vdash n} \prod_{i=1}^{|\vec{\lambda}|} R_{\lambda_i}.$$

For example, for $n = 1$, only $\vec{\lambda} = (1)$ is a composition for 1, so the RHS is just $R_1 = 1$, so $R_2 = 1$. And then for $n = 2$, we only have two possibilities $\vec{\lambda} = (1, 1), (2)$, and so the RHS is

$$R_2 := R_1 R_1 + R_2 = 1 + 1 = 2.$$

- Calculate R_3 .
 - Calculate R_4, R_5 .
3. Define a *partition* $\lambda \vdash n$ to be a **multiset** (multiset means an element can appear multiple times, but order does NOT matter, unlike in compositions!) $\lambda = \{\lambda_1, \dots, \lambda_k\}$ where $\lambda_1 + \dots + \lambda_k = n$ and $\lambda_i \in \mathbb{N}^+$, and k is the length of λ , denoted $k = |\lambda|$. For example for $n = 3$, the partitions of n are $\{1, 1, 1\}, \{2, 1\}, \{3\}$. Because order does not matter, usually we choose a “canonical ordering”, namely we write everything in weakly decreasing order: $\lambda_1 \geq \dots \geq \lambda_k$
- List out all partitions $\lambda \vdash 4$ and 5.
 - Can you find a pattern/recursive formula?